

Firm Debt Relief in Financial Downturn

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Can targeting debt relief improve stabilization policy in a financial crisis?

Firm debt relief is more common; policies vary on subsets of firms targeted:

- Paycheck Protection Program (2020-2021): Smaller firms
- Auto industry bailouts (2008-2009): “Big 3” in US auto

Differing economic support:

- Smaller, younger firms rely more on debt for investment (Faff et al., 2016)
- Large firms account for greater changes in aggregates (Crouzet & Mehrotra, 2020)
- ...while generating a vast literature [▶ Lit.](#)

Can targeting debt relief improve stabilization policy in a financial crisis?

- ▷ How should we build these targets?

Answering the Question

Non-linear general equilibrium model with firm heterogeneity and financial frictions

Heterogeneity:

- Analyze targeting different subsets of firms
 - Shape of distribution matters for aggregation outcomes
 - Unconditional size distribution matching U.S. firms
 - *Untargeted* age-size distribution of U.S. firms
- ▷ How: Persistent idiosyncratic productivity from bounded Pareto

Financial friction:

- Collateralized borrowing → slow growth for young, small firms
 - Capital misallocation
- ▷ Crisis: Shock to collateral constraint

Examined Debt Relief Policies

Debt relief policy where government pays a fraction of firm debt:

1. Targeting largest excess return to investment
 - ▷ Expected discounted marginal benefit of capital investment minus cost
 - Effective policy, may not be readily available to policymakers
2. Size-targeted policy
 - ▷ Small, medium, and large firms
 - Measures readily available to policymakers; historical precedents
3. Age-targeted policy
 - ▷ Young, middle-age, mature firms
 - Time permitting

Firms

Firm state (k, b, ε, a) : capital, debt, idiosyncratic productivity, age

- DRS production technology: $z\varepsilon F(k, n)$
- Enter period with ε ; retain with probability ρ_ε
 - Probability $(1 - \rho_\varepsilon)$ draw new ε from bounded Pareto distribution
- Age dependent exit shock: $\pi_d(a)$
 - Considerations for age-based policies
- Intertemporal decisions on k' and b'
 - Collateralized debt limit: $b' \leq \zeta k$ and loan discount factor: $q(S)$
 - Debt relief loosens budget $\rightarrow$ afford more k'

Aggregate state: $(\zeta, z, \mu, \theta) = S$

► Budget + Dividends

Government

- Total current-period debt relief: $\int g(k, b, \varepsilon, a; S) b \mu(d[k \times b \times \varepsilon \times a])$
- Borrows: θ' at risk-free rate
- Levies payroll tax: $\tau(S)$ when paying outstanding debt obligations: θ

Budget constraint:

$$\tau(S)w(S)N(S) + q(S)\theta' \geq \theta + \int g(k, b, \varepsilon, a; S) b \mu(d[k \times b \times \varepsilon \times a])$$

Policy funded by government debt; when repayment begins, $\tau(S)$ must also satisfy fiscal rule:

$$\theta' = (1 - \phi)\theta$$

- ϕ is the fraction of public debt paid per period

► Determining $\tau(S)$

Size & Age/Size Distributions

Measured by Employment (Business Dynamics Statistics: 1990-2006)

Pareto bounds, $[0.497, 0.937]$, and shape, (5.5) , targeting unconditional size distribution

Employment Bins	Emp. Share	Pop. Share	
		BDS	Model
Small (1-19)	0.201	0.885	0.880
Med. (20-499)	0.319	0.112	0.101
Large (500+)	0.480	0.003	0.019

Untargeted age-size distribution:

Age	0	1	2	3	4	5	Avg.
BDS:	0.260	0.338	0.378	0.417	0.451	0.477	0.368
Model:	0.260	0.302	0.362	0.427	0.506	0.595	0.377

Debt Relief in a Crisis

Drop in ζ , remain low for 4 periods, delayed recovery (Khan & Thomas, 2013) ► ζ_{low}

Common across all policies:

- Relief occurs on impact; total size of policy held constant at $4\% Y_{ss}$
 - PPP loan forgiveness was roughly 3.7% of U.S. real GDP by 2023
- Taxes increase in period 7 to service public debt
 - Half-life of output recovery

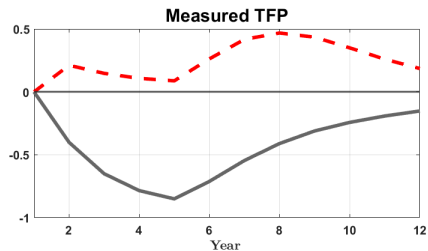
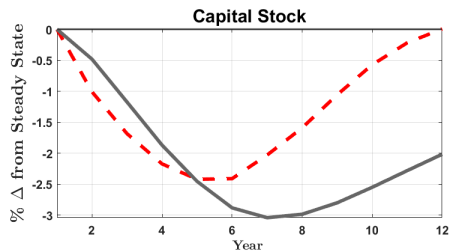
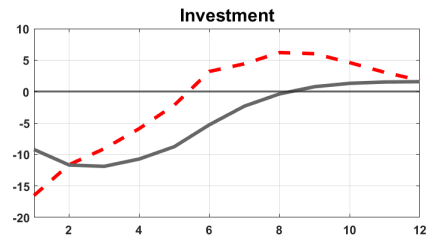
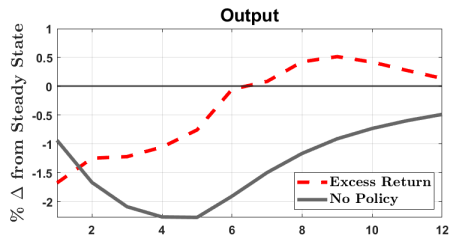
Excess return policy: Reduce/equalize largest levels of excess return from top stair down

- Continue until policy funds exhausted

Size-based policy: Eligibility based on model population shares

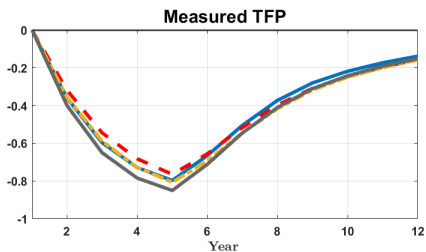
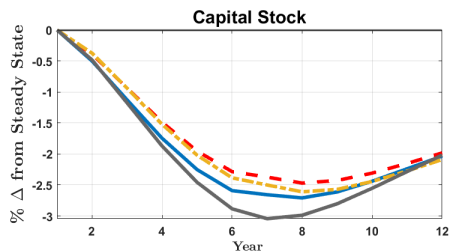
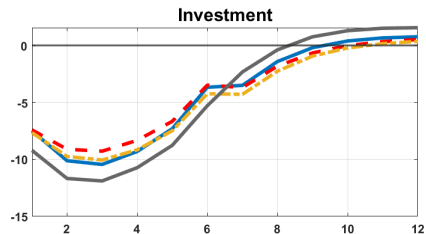
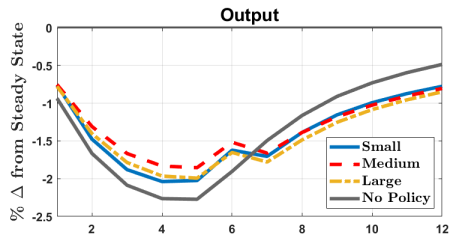
- $g(\cdot)$ varies across groups to maintain 4% cost

Policy Targeting Highest Excess Return Levels



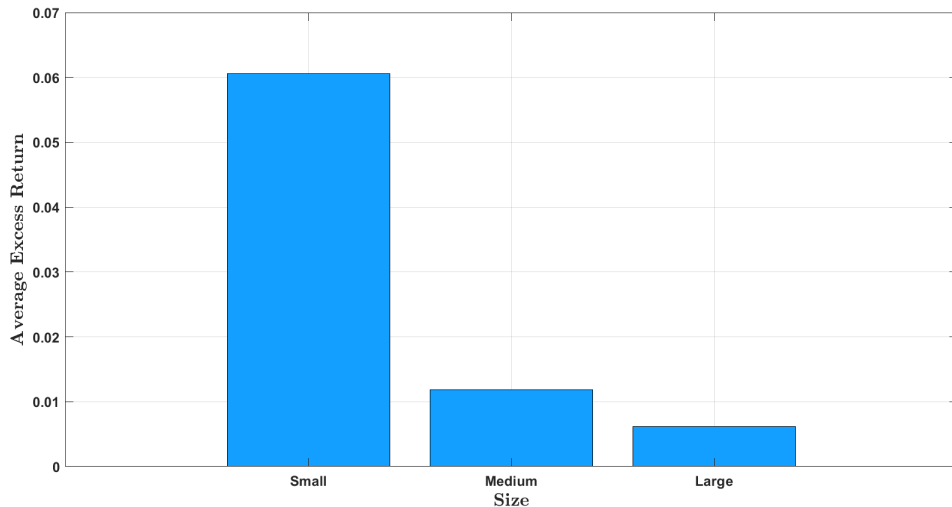
► Additional series

Policy Targeting Small, Medium, Large Firms



► Additional series

Average Excess Return Across Firm Size



Conclusion

Non-linear model of heterogeneous firms and financial frictions, matching the unconditional size, and age-size, distributions of U.S. firms to study targeted debt relief in financial crisis

I consider policies targeting firms by excess return to investment, size, and age:

- Excess return policy outperforms all others
 - Fall in output diminished by 26% compared to no policy
- Among remaining policies, targeting medium size firms reduces fall in aggregates most
 - More likely to become large economic players than small firms
 - Further from their efficient investment than larger firms

Future work: alternative tax plan to mitigate dip, heterogeneity in new ε arrival probabilities

Appendix

Policy Targeting Young, Middle Age, Mature Firms

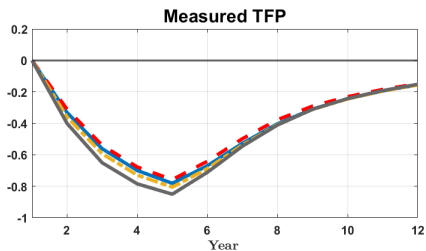
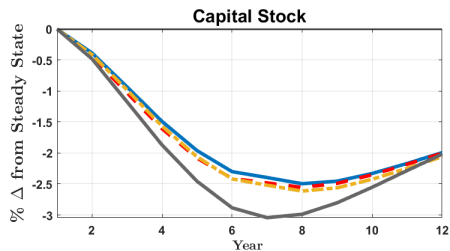
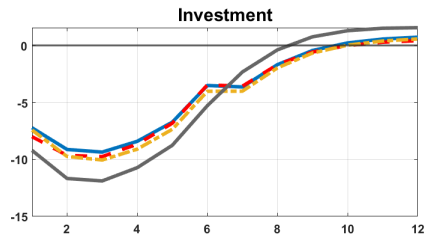
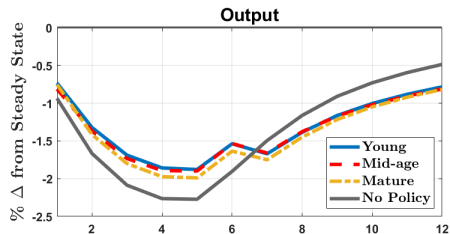
3 experiments for young, middle age, mature firm policy targets

- Age bins: $[0 - 5]$, $[6 - 10]$, $11+$

In order to keep total size of policy constant, relief per firm must vary across policies:

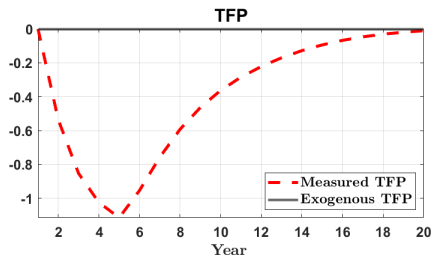
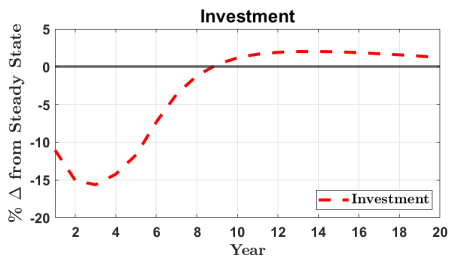
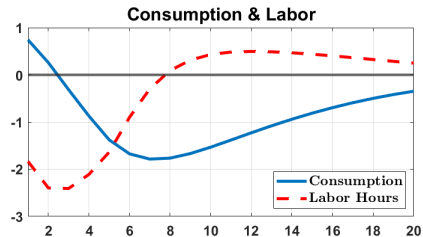
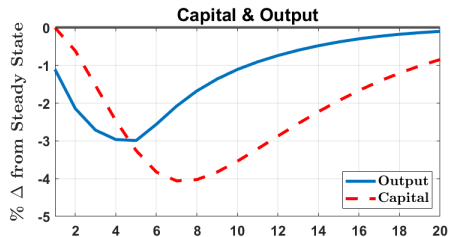
- Fraction of b paid for **young** firms: 0.352
- Fraction of b paid for **middle age** firms: 0.263
- Fraction of b paid for **mature** firms: 0.071

Policy Targeting Young, Middle Age, Mature Firms



► Additional series

Response to Credit Crisis: No Policy



Firm Budget Constraint & Debt Relief

Budget:

$$D \leq x(k, b, \varepsilon, a; S) - k' + q(S)b'$$

Cash:

$$x(k, b, \varepsilon, a; S) = z\varepsilon F(k, n) - (1 + \tau(S))w(S)n(k, \varepsilon; S) + \\ (1 - \delta)k - (1 - \mathcal{J}(b)g(k, b, \varepsilon, a; S))b$$

where: $\mathcal{J}(b) = \begin{cases} 1 & \text{if } b > 0 \\ 0 & \text{if } b \leq 0 \end{cases}$

► back

Firm Problem

Start of Period Value:

$$V_0(k, b, \varepsilon, a; S_I) = \underbrace{\pi_d(a) x(k, b, \varepsilon_i, a; S_I)}_{\text{exiting value}} + (1 - \pi_d(a)) \underbrace{V(k, b, \varepsilon, a; S_I)}_{\text{continuation value}}$$

Continuation Value:

$$V(k, b, \varepsilon_i, a; S_I) = \max_{k', b', D} \left[D + \sum_{m=1}^{N_s} \pi_{I,m}^S d_m(S_I) \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon V_0(k', b', \varepsilon_j, a'; S'_m) \right]$$

subject to:

$$0 \leq D \leq x(k, b, \varepsilon_i, a; S_I) + q(S_I)b' - k'$$

$$b' \leq \zeta k$$

$$\mu' = \Gamma(S_I)$$

Household Problem

Representative HH maximizes lifetime value, $W(\lambda, \kappa; S)$

- Chooses: consumption, c , labor, n^h , shares, λ' , bond holdings, κ'

$$W(\lambda, \kappa; S) = \max_{c, n^h, \lambda', \kappa'} \left[U(c, n^h) + \beta W(\lambda', \kappa'; S'_m) \right]$$

$$\text{st: } c + q(S_I)\kappa' + \int \rho_1(k', b', \varepsilon', a'; S'_m) \lambda'(d[k' \times b' \times \varepsilon' \times a']) \leq$$

$$w(S)n^h + \kappa + \int \rho_0(k, b, \varepsilon, a; S_I) \lambda(d[k \times b \times \varepsilon \times a])$$

- ρ_1 is ex-dividend price of a share, ρ_0 is dividend-inclusive value of a share, $\Gamma(\mu)$ is known

► back

Representative Household

- Period utility $U(C, 1 - N)$, and discount factor $\beta \in (0, 1)$
- Supplies: labor for wage $w(S)$, loans at risk free rate $q(S)^{-1}$
- Implied restrictions for equilibrium prices:
 - $w(S) = \frac{D_2 U(C, 1 - N)}{D_1 U(C, 1 - N)}$
 - $d_m(S_l) = \beta \frac{D_1 U(C'_m, 1 - N'_m)}{D_1 U(C, 1 - N)}$
 - $q(S_l) = \beta \sum_{m=1}^{N_s} \pi_{l,m}^s \frac{D_1 U(C'_m, 1 - N'_m)}{D_1 U(C, 1 - N)}$
- Equilibrium decision rules $C = C(S)$, $N = N(S)$

► HH Problem

► Calibration

Determining ζ_{low}

33% drop in ζ , remains low for 4 periods, recovers 31.25%/year (Khan & Thomas, 2013)

- 26% fall in borrowing in the model, matching fall in C&I loans 2008-2011
- Supply shock in loanable funds market similar to 2008 (Duchin et al., 2010)

▶ back

Comparison to Untargeted Policy

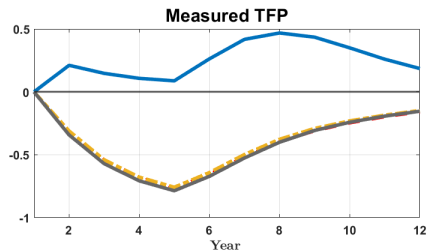
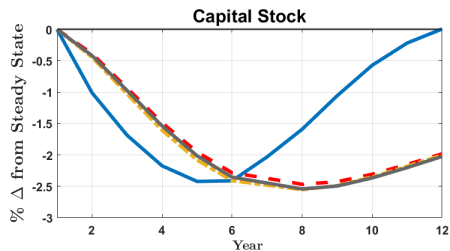
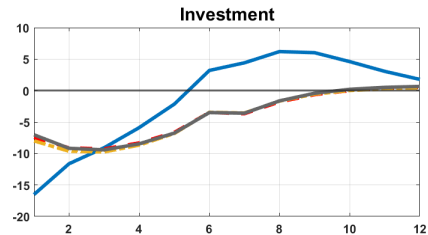
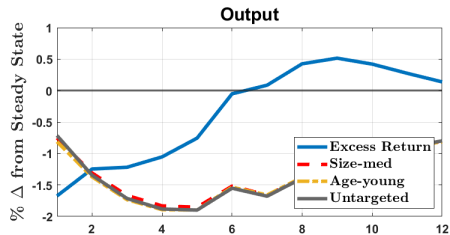
I consider one final policy: all indebted firms are eligible

- Fraction of total firm debt relieved: 0.047

Highlight effects of diminishing marginal returns

- Targeted policy can focus on key variables
- But, more concentrated policy → stronger diminishing marginal returns effects

Untargeted Policy Compared to Best Alternative Policies



Literature

- Fiscal policy to alleviate financial downturn:
 - Bianchi (2016), Jeanne, & Korinek (2020), Elenev, Landvoigt & Van Nieuwerburgh (2022), Angeletos, Collard, & Dellas (2023)
 - ▷ **My contribution:** Value of policy resources varies over nontrivial distribution of firms
- Distributional effects of policy intervention:
 - Guner, Ventura, Xu (2008), Buera, Moll, & Shin (2013), Gourio and Roys (2014), Jo & Senga (2019)
 - ▷ **My contribution:** Extend distributional analysis to the topic of debt relief in a crisis
- Transmission of shocks from financial conditions:
 - Kiyotaki & Moore (1997), Jermann & Quadrini (2012), Khan & Thomas (2013), Jo (2024)
 - ▷ **My contribution:** Analyze how debt relief may improve such conditions

Market Clearing

$$\text{Capital : } K = \int k \mu(d[k \times b \times \varepsilon \times a])$$

$$\text{Labor : } N = \int n(k, \varepsilon) \mu(d[k \times b \times \varepsilon \times a])$$

$$\text{Output : } Y = \int z \varepsilon F(k, n(k, \varepsilon)) \mu(d[k \times b \times \varepsilon \times a])$$

$$\text{Firm-debt : } B = \int (b | b > 0) \mu(d[k \times b \times \varepsilon \times a])$$

$$\begin{aligned} \text{Consumption : } C = & Y - (1 - \pi_d(a)) (K' - (1 - \delta) k) \\ & + \pi_e ((1 - \delta) k - k_0) \end{aligned}$$

Distribution

The distribution of firms is denoted by measure μ , defined on the Borel algebra, $\mathcal{S}$, generated by the open subsets of the product space, $\mathbf{S} = \mathbf{K} \times \mathbf{B} \times \mathbf{E} \times \mathbf{A}$.

$\forall (A, \varepsilon_j) \in \mathcal{S}$ defines Γ , where $\chi(k_0) = \{1 \text{ if } (k_0, 0) \in A; 0 \text{ otherwise}\}$

$$\begin{aligned} \mu'(A, \varepsilon_j) = & \\ (1 - \pi_d(a)) \int_{\{(k, b, \varepsilon_i, a) | (g^K(k, b, \varepsilon_i, a; s, \mu), g^B(k, b, \varepsilon_i, a; s, \mu)) \in A\}} & \pi_{ij} \mu(d[k \times b \times \varepsilon_i \times a]) \\ + \pi_e \chi(k_0) H(\varepsilon_j) & \end{aligned}$$

Constrained and Unconstrained Investment

Collateral constraint binds to varying degrees across firms

Non-binding: $k'(k, b, \varepsilon, a; S) = k^*(\varepsilon; S)$

- Debt relief provides no extra investment
- Additional resources saved, $b' = \frac{k^*(\varepsilon, S) - x(k, b, \varepsilon, a; S)}{q(S)}$
 - Does offer protection from future binding constraint

Binding: $k'(k, b, \varepsilon, a; S) = x(k, b, \varepsilon, a) + q(S) \underbrace{\zeta k}_{b' = \zeta k} < k^*(\varepsilon; S)$

- Debt relief for these firms increases investment, reduces misallocation

Measuring Investment Inefficiency

Excess return to investment: expected discounted marginal value of investing minus cost

$$\mathbb{E}_{\pi^S} \mathbb{E}_{\pi^\varepsilon} \left[d_m(S_l) \left(\frac{\partial \pi(k', b', \varepsilon_j, a'; S'_m)}{\partial k'} + (1 - \delta) \right) \right] - 1$$

where π is profit ▶ Parameterization

With efficient investment, excess return is 0

The further $k'(k, b, \varepsilon, a, S)$ is from $k^*(\varepsilon, S)$, the higher the return

Excess Return to Investment

Given the parameterization of the model, marginal benefit of capital investment should equal marginal cost of 1:

$$\frac{\alpha}{1-\nu} \frac{\beta}{p(S)} \left(\sum_{m=1}^{N_s} \pi_{l,m}^s p(S'_m) \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon k'^{\frac{\alpha+\nu-1}{1-\nu}} \left[z_m \varepsilon_j \left(\frac{\nu z_m \varepsilon_j}{(1+\tau(S)) w(S)} \right)^{\frac{\nu}{1-\nu}} - (1+\tau(S)) w(S) \left(\frac{\nu z_m \varepsilon_j}{(1+\tau(S)) w(S)} \right)^{\frac{1}{1-\nu}} \right] + (1-\delta) \right) = 1$$

where, $p(S) = \frac{\partial U(C, 1-N)}{\partial C}$

With insufficient investment, LHS > 1

[▶ back](#)

Determination of Dividends

Final decision for continuing firms is their dividend payout

Constrained firms choose $D = 0$

- Greater value in investing those resources since excess return > 0

Unconstrained firms face one of two cases:

1. Positive probability of future binding collateral constraint:
 - Direct incentive to save: choose $D = 0$
2. No possibility of future binding collateral constraint:
 - *Minimum saving policy*; issue just enough dividends to remain unconstrained in all states

▶ Min. Savings Sol.

Minimum Savings Policy

Define:

$$B(k, \varepsilon; S) \equiv \min_{\{\varepsilon_j | \pi_{ij} > 0 \text{ and } S_m | \pi_{im}^S > 0\}} \tilde{B}(k^*(\varepsilon), \varepsilon_j; S'_m)$$

where,

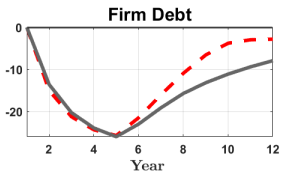
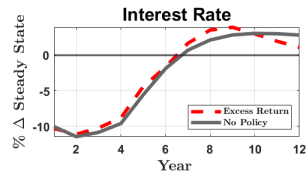
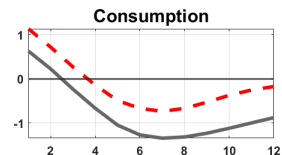
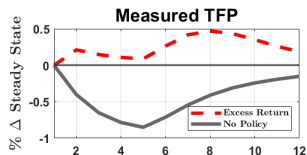
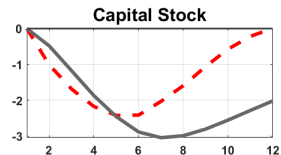
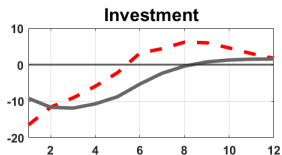
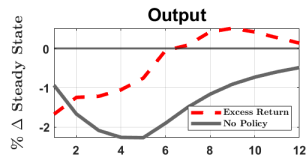
$$\begin{aligned} \tilde{B}(k, \varepsilon; s, \mu) = & z\varepsilon F(k, n^*(k, \varepsilon)) - (1 + \tau(S))w(S)n^*(k, \varepsilon) \\ & + q(S) \min \{B(k, \varepsilon; S), \zeta k\} - (k^*(k, \varepsilon) - (1 - \delta)k) \end{aligned}$$

and finally, dividends are:

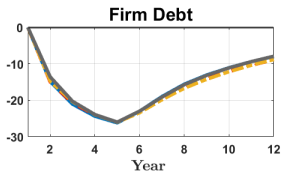
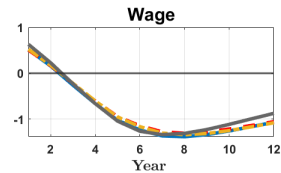
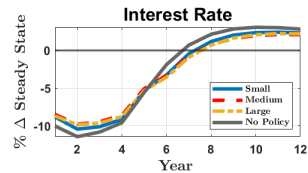
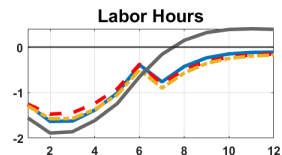
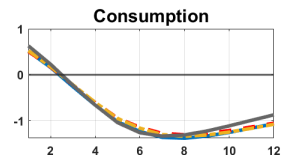
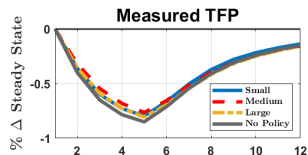
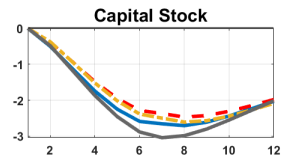
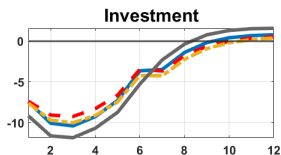
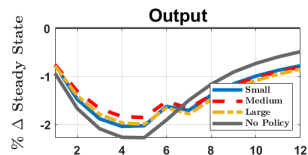
$$D(k, b, \varepsilon, a; S) = x(k, b, \varepsilon; S) - k^*(k, \varepsilon) + q(S) \min \{B(k, \varepsilon; S), \zeta k\}$$

▶ back

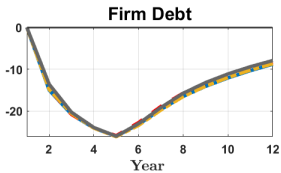
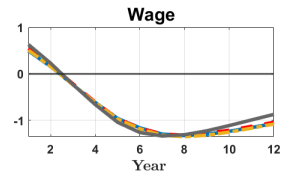
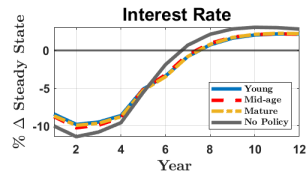
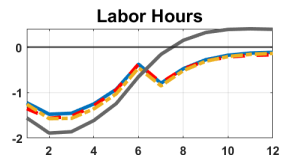
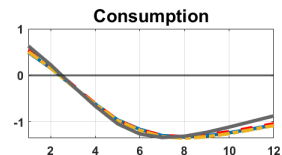
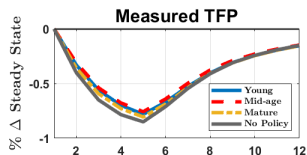
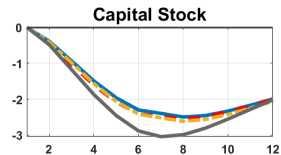
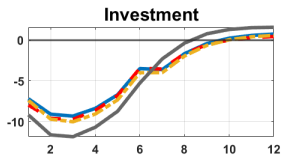
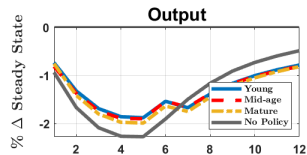
Extra Series - Excess Return Target



Extra Series - Size Targets



Extra Series - Age Targets



Annual Calibration

$$U(C, 1 - N) = \ln(C) + \psi(1 - N) \quad z\varepsilon F(k, n) = z\varepsilon k^\alpha n^\nu$$

$$k_0 = \chi \int k \mu(d[k \times b \times \varepsilon \times a])$$

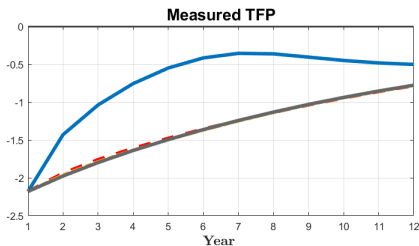
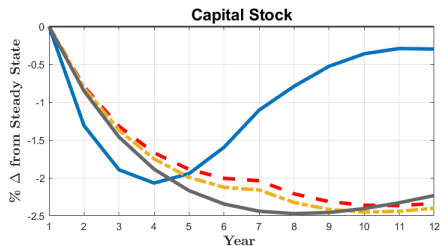
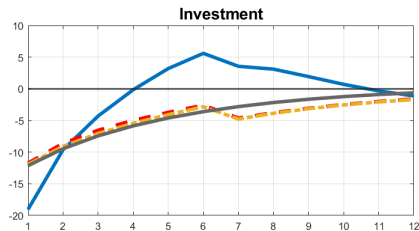
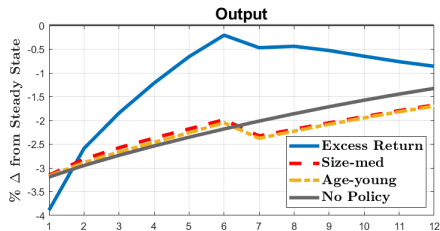
► HH Conditions

Parameter			Target		Model
β	discount factor	= 0.960	real interest rate	= 0.040	0.041
ψ	leisure preference	= 2.140	labor hours	= 0.333	0.332
ν	labor share	= 0.600	labor share	= 0.600	0.600
δ	depreciation	= 0.069	investment/capital	= 0.069	0.069
$\frac{b_0}{k_0}$	entrant leverage	= 0.400	entrant leverage	= 0.400	0.400
α	capital share	= 0.280	capital/output	= 2.300	2.305
χ	fraction of entrant K	= 0.208	avg. n_0/N	= 0.260	0.260
ρ_ε	maintain ε	= 0.990	std dev. i/k	= 0.337	0.358
ζ_0	collateral fraction	= 0.981	debt/assets	= 0.372	0.372

► back

► Exit Rates

Selected Policies Following TFP Shock



- 2% shock with persistence of 0.909

Firm Exit Rates by Age (BDS, 1990-2006)

Age:	1	2	3	4	5	6-10	11+
$\pi_d(a)$:	0.2478	0.1640	0.1356	0.1174	0.1062	0.0840	0.0655

Firm entrant rate = 10.6%

- Entrant rate selected to keep mass of firms at 1

[▶ back](#)

Perfect Foresight Solution -1

1. Guess a vector of $\{\hat{\tau}\}_0^{T+1}$ and $\{\hat{C}\}_0^{T+1}$
 - τ' is needed for k' decision
 - $\hat{C}$ implies a $w(S)$ and $q(S)$
2. Back-solve decision rules from date T
3. Forward-solve the distribution (and find aggregates) for each T
4. Back out $\tilde{C}$ implied by aggregate resource constraint
5. Solve for $\tilde{\tau}$ by following:

Determining $\tau(S)$

Define: $Balance \equiv \hat{\tau}(S)w(S)N(S) + q(S)\theta' - \theta - T(\Theta, S)$

Then, $\hat{\tau}(S)w(S)N(S) = Balance + q(S)\theta' - \theta - T(\Theta, S)$

Define: $\Delta\hat{\tau}(S)$ as change in $\hat{\tau}(S)$ such that:

$$(\hat{\tau}(S) + \Delta\hat{\tau}(S))w(S)N(S) = \theta + T(\Theta, S) - q(S)\theta' \quad \text{and} \quad \theta' = (1 - \phi)\theta$$

- This is the increase in $\hat{\tau}(S)$ needed to set $Balance = 0$, while the fiscal rule is satisfied

Then: $\Delta\hat{\tau}(S) = \left((\theta + T(\Theta, S) - q(S)\theta') \left(\frac{1}{w(S)N(S)} \right) \right) - \hat{\tau}(S)$

So, $\tilde{\tau} = (\hat{\tau}(S) + \Delta\hat{\tau}(S))$

6. Check guess, set $\{\tilde{\tau}\}_0^{T+1} = \{\hat{\tau}\}_0^{T+1}$ and $\{\tilde{C}\}_0^{T+1} = \{\hat{C}\}_0^{T+1}$, and repeat govcrisis

► back